



PART A Master Technical Production Manual

Production basis, systems, lighting, sound, projection, effects, the 83-cue show run, procedures, recommendations and worksheets — for the R13.1 show.

Revision R13.1 · 28 September 2026

PLANTAGENET HALL · REVISION R13.1 · 28 SEPTEMBER 2026

Master Technical Production Manual

Lighting · Sound · Projection · QLab · LSC Mantra · StudioLive · Haze · Stage management · Effects · Show control

System	Project configuration
Production basis	Licensed script, Nick Lawrence Pantomimes, revised April 2026 (55 pp as supplied); casting sheet 28 Sep 2026
Venue	Plantagenet Hall — installed LED rig, FOH truss, two stage bars, floor booms, projector and screen
Lighting control	LSC Mantra Lite + 2 wings · TLM_SHOW_2026_R13_FLASHY_SCENE_SPLIT.mtr
Show control / media	QLab 5 · TLM_Show_R13_1.qlab5 — one GO fires light, sound and video
Audio	PreSonus StudioLive 16 · radio mics + QLab stereo return ch 11–12
Effects	Hazer #40, optional mirror ball + pinspots, water pistols where approved

IMPORTANT STATUS

A production planning, programming and operating manual. Fixture modes, rigging loads, power, projector throw, fire-alarm interaction and final stage dimensions must be verified on site. It is not a structural, rigging, electrical or fire-engineering certification.

Document control

Item	Detail
Revision	R13.1 — the R13 FLASHY SCENE SPLIT show with the R13.1 fixes (supersedes R8/R9)
Date	28 September 2026
Show files	TLM_Show_R13_1.qlab5 · TLM_SHOW_2026_R13_FLASHY_SCENE_SPLIT.mtr · folder TLM_R13_REBUILT_Show_Files
Companion parts	B Lighting · C Patch · D Mantra · E QLab · F Cue sheets · G DSM script · H Props · I Projection · J Prompt copy · K Quick reference · L Stream Deck & tech test

What changed in R13.1 (since R8)

- **Show control rebuilt (R13):** the Mantra show is split — P2 Act One scenes, P3 Act Two scenes, P4 songs, P5 FX, P6–P7 look library, P8 the full 153-position backup. QLab targets the exact memory and cue for every step.
- **Colour-forward lighting (R13):** stronger LX1 colour, side light, LX2 backlight and PixBar movement over neutral faces; fast colour snaps on musical impacts; no strobe (Part B).
- **Songs:** ten song memories (S1–S10) with 66 section cues; nine tracks supplied (S2, S3, S5 still vocal versions; S7 to choose).
- **R13.1 fixes:** flash returns fired by QLab (no desk link times), memory releases so looks can't stack, HAYWIRE on at Q57 and a new **Q57b** Ariel restored, 58 audio cues re-targeted to the new SFX pack, Q19.2 rain burst, song fade-and-stops, stale R11 aliases cleared.
- **Projection:** backdrops R10 — 21 in the cue plan, ship scenes matched to the script, seamless 15–16 s loops (Part I).
- **Script:** re-paged to the licensed script as supplied (55 pp); the contents now list a Scene Eleven that has no text (R-01).
- **Casting:** microphone plan and beginners built from the casting sheet (S8).

Key: teal boxes = recommendations (R-xx), coral = verify on site / safety, navy = operating rules, yellow = new in R13.1.

R Recommendations register

16	13	2	4			
high priority	medium priority	low priority	new in R13.1 (R-28 to R-31)			
ID	Priority	Area	Recommendation	Owner	Ref	Done
R-01	High	Script	Scene Eleven is in the contents but not the text; confirm with director/licensor	SM / Director	\$12	☐
R-02	High	Lighting	Use the venue's 12 Lightsky profiles for face light and specials	LX / TD	\$6	☐
R-03	High	Show control	One cue language across every system (QLab number = master cue; desk names carry it)	SM / all ops	\$7	☐

ID	Priority	Area	Recommendation	Owner	Ref	Done
R-04	High	Safety	Flash, haze and loud-effect warnings; single hits only, ≤3 flashes/s — FOH notice supplied	TD / FOH	\$10	☐
R-05	High	Electrical	RCD protection and test-and-tag for all show power	TD / venue	\$5	☐
R-06	High	Audio	Wireless mic frequencies legal and coordinated	Sound op	\$8	☐
R-07	High	QLab	Harden the show Mac	QLab op	\$7	☐
R-08	High	Redundancy	Emergency playback rig on StudioLive ch 13–14	Sound / QLab op	\$16	☐
R-09	Medium	Audio	Wireless count 8 + 1 spare minimum	Sound / producer	\$8	☐
R-10	Medium	Lighting	Keep the venue's two-universe patch	LX	\$6	☐
R-11	Medium	Projection	Projector centre stage towards the back: throw, shadows, signal	Projection	\$9	☐
R-12	Medium	Stage	Make blackout scene changes safe	SM / deck	\$11	☐
R-13	Medium	Haze	Written haze plan with the venue	TD / venue	\$10	☐
R-14	Medium	Rigging	Secondary safety on the mirror ball	Venue / TD	\$10	☐
R-15	Medium	Rehearsal	Paper tech and dry tech	SM / TD	\$14	☐
R-16	Medium	Comms	Two-channel headset plan	SM / TD	\$2	☐
R-17	Medium	Network	Fixed 2.0.0.x addresses on one wired show network	TD	\$3	☐
R-18	Medium	Crew	Name a Deck/ASM and a FOH manager	Producer	\$2	☐
R-19	Low	Audience	Consider a relaxed performance	Producer / Sound	\$8	☐
R-20	Low	Documentation	Photograph each look at dress	LX	\$6	☐
R-21	High	QLab	Licence QLab — R13.1 fires every lighting cue from QLab	Producer / QLab op	\$7	☐
R-22	High	Rigging	Rate the projector bar and bond the projector	Venue / TD	\$9	☐
R-23	High	Safety	Water pistols are scripted to hit the audience	Director / TD / FOH	\$11	☐
R-24	High	Rigging	Two stage bars and side booms: loads, bases, wing space	Venue / TD	\$4	☐
R-25	Medium	Lighting	Turn the upstage bar away from the projection screen	LX	\$4	☐
R-26	High	Rigging	Floor booms in place of side trusses	TD / producer	\$4	☐
R-27	High	Crew	One show operator: keep the sound mix simple	Producer / Sound	\$2	☐
R-28	High	Audio	NEW: supply the backing tracks (S2, S3, S5, check S4), the S7 song and the ten silent placeholders before dress	MD / Sound	\$8	☐
R-29	High	Show control	NEW: set the show-Mac QLab settings (audio, video, network interface) and add the Mantra OSC trigger	QLab op / LX	\$7	☐
R-30	Medium	Lighting	NEW: check the R13 colour-forward looks from the house — faces, screen washout, bold backlight choices	LX / Director	\$6	☐
R-31	Medium	Show control	NEW: agree who calls the 56 song-section GOs (DSM or QLab op to the music)	SM / QLab op	\$12	☐

1 Production basis and technical philosophy

Fixed lighting and reliable local fallback rather than moving-head dependence or fragile automation. The projector gives scenery and motion, the fixed rig lights performers, QLab plays media and fires the Mantra, the StudioLive mixes microphones. Haze and the mirror ball are selective effects.

Requirement	Technical response
No moving heads	Zone washes and pre-focused specials; crossfade specials to suggest movement (voice transfer)
Projection screen	Front/key beams kept off the screen; projection for environment, lighting for performers
Colour-forward R13 design	Energy from LX1, side, back and PixBars; FOH faces stay near-white
One show operator	QLab GO fires light, sound and video together; Mantra and StudioLive beside the Mac as fallback
Haze	Short, controlled bursts for underwater, storm, magic and lair; clear air for dialogue

OPERATING PRINCIPLES

- 1 Actor-paced cues are manual GOs — never hard-timed to a line.
- 2 Mantra is the lighting master: the show must stay runnable from the desk (P2/P3/P4 memories).
- 3 StudioLive is the live-audio master; QLab playback is separate from live mics.
- 4 Projection is enhancement: every scene must read with lighting alone.
- 5 Safety and visibility first: stop water, haze or flash effects if conditions become unsafe.
- 6 The same cue numbers in the prompt book and every operator sheet (R-03).

2 Crew structure and communications

Role	Responsibility	Comms
Technical Director	Integration, safety sign-off, patch, backups, technical decisions	B
Stage Manager (DSM)	Prompt book (Part J), standbys/GOs (Part G), scene changes, emergency calls	A + B
Show operator (QLab + desks)	GO in QLab (fires light, sound, video); Mantra and StudioLive fallback	A
Sound (if separate)	StudioLive mix, radio mics, spare-mic response	A
Haze / FX	Haze and effects on cue, within venue rules	A
Deck / ASM	Props, tabs, water, shell/plinth, jellyfish, drying and resets (R-18)	B
FOH manager	Audience safety, house-light moments (Q13, Q64), sensory warnings (R-04)	B / phone
Call	Meaning	
STANDBY	Operator prepares the cue and confirms ("QLab standing by")	
GO	Execute the called cue now	
HOLD	Do not execute; keep the current safe state	
GO LX ONLY / GO SOUND ONLY	Execute that part only	
STOP PLAYBACK	Stop QLab playback without muting live mics	

R-16 • R-18 • R-27

Channel A carries show calls, Channel B tech/backstage. Name a Deck/ASM and a FOH manager. With one show operator, keep the sound mix simple: principals on one DCA, playback on another.

3 System architecture, signal flow and network

From	To	Connection / purpose
QLab Mac	Mantra Lite (2.0.0.1)	OSC /PlayMemory over UDP port 8000 — every lighting cue, releases, flash returns
QLab Mac	StudioLive ch 11–12	Stereo playback return (USB)
QLab Mac	Projector	HDMI via HDBaseT/active cable to centre stage — backdrops and VID-99 black
Stream Deck	QLab Mac	USB — GO, Panic, E1–E3 hotkeys
Mantra	Fixtures	U1 DMX from the desk XLR · U2 Art-Net/sACN to a node (2.0.0.10) for the C42s
StudioLive	PA / monitors	Main L/R, Aux 1–2
Device	Address	Notes
Mantra Lite	2.0.0.1	In the show file; Art-Net + sACN on (turn off the unused one)
Art-Net/sACN node	2.0.0.10	Universe 2
QLab Mac	2.0.0.20	Manual IPv4, Wi-Fi off
Backup Mac	2.0.0.21	Optional
StudioLive (UC Surface)	2.0.0.30	Optional
Tablet / remote	2.0.0.40	Optional

FALLBACK HIERARCHY

Network fails: run lighting from the Mantra P2/P3/P4 memories. QLab fails: live mics stay on the StudioLive, lighting from the desk, songs from emergency playback (R-08). Projection fails: continue on lighting. One failure never stops the show on its own.

4 Venue, stage and lighting layout

The venue's own rig as patched: 12 Lightsky 8800-C42 profiles (#1–12), 10 Tour Pro Zooms (#13–22), 11 TourCOB PARs (#23–32, 39), 6 PixBars (#33–38) and the hazer (#40). FOH truss #1–8; LX1 #9–18; LX2 #19–20, 23–28, 33–39; SR boom #21, 29, 30; SL boom #22, 31, 32. The projector hangs centre stage towards the back and throws upstage. Rig plan, positions and focus: Part B §1 and §7; Quick Reference card 1.

R-24 • R-25 • R-26

Confirm bar loads, boom bases and wing space. Turn LX2 away from the screen — R13 runs the backlight hard. Use a weighted floor boom in each wing (no side trusses).

5 Power, data, cable and rigging

System	Arrangement	Pre-show check
Lighting power	Venue lighting circuits, RCD-protected	Circuit ID, connectors, strain relief
Audio power	Same phase/earth where possible	Hum, overload, chargers
Projector	Dedicated outlet	Warm-up, ventilation, signal, bond
Haze	Supply to rating and venue policy	Fluid, detector isolation logged
Show network	Unmanaged switch, labelled Cat6	Link lights, addresses
Compliance	In-date test-and-tag (AS/NZS 3760)	Untagged items quarantined

Keep water trajectories 2 m from joins, floor boxes, the projector and FOH control. Terminate every DMX line; never Y-split DMX. Rigging, loads and permanent wiring are for the venue or a qualified person (R-05).

6 Lighting system and LSC Mantra

Full design in Part B and programming in Part D. The R13.1 show file holds:

Page	Contents
P1	Venue / rig base — M1 STAGE WORK · M2 FULL WHITE · M3 WARM · M4 COOL · M5 BLUE · M6 RED · M9 PIXBAR WASH · M10 CURTAIN CALL
P2	Act One — scene sections — M1 Q1–11 · M2 Q12–17 · M3 ship/storm Q18–27.5 · M4 shore/duet Q28–30.5 · M5 bar/transformation Q31–37 · M6 interval Q38 · M9 WORK/FOCUS · M10 SAFE LIGHT
P3	Act Two — scene sections — M1 palace Q39–44 · M2 jellyfish Q45–49 · M3 lair/voice/haywire Q50–58.5 · M4 dry land Q59–59.7 · M5 wedding Q60–63.5 · M6 bows/end Q64–65
P4	Songs — M1 S1 Rock Lobster ... M10 S10 He's a Pirate
P5	FX / chases — M1 WARM CHASE · M2 COOL CHASE · M3 PARTY CHASE · M4 HAZE · M5 HAYWIRE · M6 JELLY PULSE
P6	Look library — M01 BLACKOUT ... M10 MAGIC
P7	Look library — M11 LAIR ... M16 BOWS · M7 = M17 SAFE LIGHT
P8	Backup / reference — M1 = the full 153-position list (BACKUP - 01 PRESHOW ...)

Look library as programmed

Swatch order: **Face** (FOH #1–6) · **Top** (LX1 #13–18) · **Side** (booms) · **Back** (LX2 COB) · **Pix** (PixBars) · **Spec** (specials). Colour as programmed in the R13.1 show file, darker = lower level, black = off.

Look	Desk	Layers	As programmed
M01 BLACKOUT	P6 M1		all out
M02 HOUSE	P6 M2		Back 10% blue
M03 UNDERWATER	P6 M3		Face 55% cool white · Top 55% cyan · Side 60% lime · Back 88% red
M04 COMEDY	P6 M4		Face 85% warm white · Top 72% hot pink · Side 68% violet · Back 62% sky blue
M05 ROYAL	P6 M5		Face 75% warm white · Top 65% gold · Side 60% hot pink · Back 82% violet
M06 SHIP	P6 M6		Face 70% warm white · Top 58% blue · Side 55% teal · Back 80% orange
M07 SHORE	P6 M7		Face 75% orange · Top 62% cyan · Side 58% lime · Back 66% red
M08 STORM	P6 M8		Face 35% cool white · Top 28% violet · Side 48% blue · Back 100% teal · Pix 72% warm white + lime
M09 LIGHTNING	P6 M9		Face 100% cool white · Top 100% sky blue · Side 100% cyan · Back 100% gold · Pix 100% sky blue + red
M10 MAGIC	P6 M10		Face 45% deep blue · Top 60% teal · Side 66% orange · Back 92% magenta · Pix 82% lime + violet
M11 LAIR	P7 M1		Face 30% cool white · Top 34% violet · Side 46% sky blue · Back 88% green
M12 VOICE SHELL	P7 M2		Face 5% cool white · Side 16% lime · Back 18% red
M13 JELLYFISH	P7 M3		Face 65% warm white · Top 68% hot pink · Side 70% violet · Back 80% sky blue · Pix 78% magenta + cyan
M14 DRYLAND	P7 M4		Face 80% warm white · Top 60% orange · Side 52% magenta · Back 56% warm white
M15 WEDDING	P7 M5		Face 90% warm white · Top 78% gold · Side 70% hot pink · Back 88% violet

Look	Desk	Layers	As programmed
M16 BOWS	P7 M6		Face 100% warm white · Top 88% magenta · Side 82% sky blue · Back 90% cyan
M17 SAFE LIGHT	P7 M7		Face 70% white

R-30 · NEW

The R13 looks are deliberately bold. At the focus session check every look from the house: faces readable, screen not washed out, and bold choices (e.g. coral-red backlight in M03 UNDERWATER) agreed with the director. Photograph each look at dress (R-20).

Before the show	During the show	If something goes wrong
Load the show; fire E3 SAFE LIGHT and one cue per memory from QLab	Single operator: GO in QLab on the SM's call fires the lighting	QLab/network down: run the matching P2/P3/P4 memory with Next Cue; press Next Cue again after each flash. Emergency: SAFE LIGHT P2 M10. Never replay a missed flash.

7 QLab programming and show control

Full guide in Part E. The workspace is already built: 479 cues — 83 master cue groups, 10 song groups and E1–E3. Cue number = master cue number; Mantra desk cues carry the same number in their names (R-03).

- ☐ Dedicated show user; sleep, screen saver, notifications, updates off
- ☐ Show folder on the Desktop with its exact name
- ☐ Audio patch → StudioLive, video stage → projector, network interface → Ethernet (R-29)
- ☐ E1–E3 hotkeys F15/F13/F14
- ☐ Wi-Fi and Bluetooth off
- ☐ Licence active (R-21)
- ☐ Mantra OSC Play Memory trigger on port 8000 (R-29)
- ☐ Show Mode for every performance

R-21 · R-29

R13.1 fires every lighting cue from QLab, so the licence is essential. The audio device, video screen and network interface can't be stored in the file — set them on the show Mac, and add the OSC remote trigger on the desk.

8 Sound system and StudioLive

Input plan (from the casting sheet)

Ch	Source	Method	Notes
1	Ariel (Hayley)	Headset/radio	Lead vocal: Part of Your World, Time of My Life; offstage voice lines (Scene Eight)
2	Prince Evan (Jett)	Headset/radio	Sea Shanty, Time of My Life
3	Octavia (Hannah)	Headset/radio	Poor Unfortunate Souls; villain dialogue
4	Dame Beluga (Michelle)	Headset/radio	Feeling Good; audience interaction
5	Willis (Eliott)	Headset/radio	Sea Shanty
6	Flanders (Kelly)	Headset/radio	Comedy lead — watch peaks
7	Queen Marina (Lorraine)	Headset/radio	Dialogue; man's voice offstage at Q56
8	Theodore / Godfrey (Emma / Jeff)	Radio, shared only if changes allow	Mute the inactive performer
9–10	Area / overhead L/R	Condensers	Sailors (Grace, Aubrey, Isabelle) in the Sea Shanty; ensemble
11–12	QLab L/R	USB return	Playback, minimal processing
13–14	Emergency playback L/R	Backup device (R-08)	Muted; unmute only on SM call
15–16	Spare / Spirits	Spare radio	Candidate: Spirit of the Sea 1/2 (Kerry, Tracey) — key speeches at Q28, Q62

Who sings what (casting sheet)

Role	Actor	Sings	Dances
Ariel	Hayley	Part of Your World · Time of My Life	Absolutely Everybody
Dame Beluga	Michelle	Feeling Good	Absolutely Everybody
Flanders	Kelly		Rock Lobster · Absolutely Everybody
Queen Marina	Lorraine		Absolutely Everybody
Octavia	Hannah	Poor Unfortunate Souls	Absolutely Everybody

Role	Actor	Sings	Dances
Theodore	Emma		Absolutely Everybody
Prince Evan	Jett	Sea Shanty • Time of My Life	Absolutely Everybody
King Godfrey	Jeff		Absolutely Everybody
Willis	Eliott	Sea Shanty	Absolutely Everybody
Slip	Flyn		Absolutely Everybody
Slap	Matthew		Absolutely Everybody
Spirit of the Sea 1	Kerry		Absolutely Everybody
Spirit of the Sea 2	Tracey		Absolutely Everybody
Jellyfish	Helen		Absolutely Everybody
Scarlotte / Sailor 3 / Courtier / Scary Fish	Grace	Sea Shanty	Rock Lobster • Crab Rave • Poor Unfortunate Souls • Absolutely Everybody
Paulette / Sailor 1 / Courtier / Scary Fish	Aubrey	Sea Shanty	Rock Lobster • Crab Rave • Poor Unfortunate Souls • Absolutely Everybody
Charlotte / Sailor 2 / Courtier / Scary Fish	Isabelle	Sea Shanty	Rock Lobster • Crab Rave • Poor Unfortunate Souls • Absolutely Everybody
Kandy / Courtier / Scary Fish	Shelli		Rock Lobster • Crab Rave • Poor Unfortunate Souls • Absolutely Everybody
Mermaid / Courtier / Scary Fish	Maddie		Rock Lobster • Crab Rave • Poor Unfortunate Souls • Absolutely Everybody
Dancer	Sarah		Crab Rave • Absolutely Everybody

Music — the ten songs

Song	At	Performed by	Track	Status
S1 Rock Lobster	Q8	MUS-A1-01	Original vocal recording	supplied
S2 Feeling Good (Austin Millz Remix)	Q14	MUS-A1-02	Backing / instrumental	vocal version - swap for backing
S3 Part of Your World	Q15	MUS-A1-03	Backing / instrumental	vocal version - swap for backing
S4 Wellerman (Sea Shanty)	Q18	MUS-A1-04	Backing / instrumental	check for vocals
S5 (I've Had) The Time of My Life	Q30	MUS-A1-05	Backing / instrumental	vocal version - swap for backing
S6 Crab Rave	Q40	MUS-A2-01	Original recording (instrumental)	supplied
S7 Jellyfish chorus (song to be decided)	Q46	MUS-A2-02	To be decided (backing, short chorus)	song still to choose (silent placeholder)
S8 Poor Unfortunate Souls	Q51	MUS-A2-03	Backing / instrumental	supplied
S9 Absolutely Everybody	Q63	MUS-A2-04	Original vocal recording	supplied
S10 He's a Pirate	Q64	MUS-FIN-01	Original recording (instrumental)	supplied

R-28 • NEW

Before dress: backing tracks for S2 Feeling Good, S3 Part of Your World and S5 Time of My Life (and S4 if the file has vocals); choose the S7 jellyfish chorus; supply house/preshow, end of Act One, interval and exit music (Q1, Q2, Q37, Q38, Q65), Ariel's recorded line (Q42.5), the cue-card underscore (Q43), the soft resolve (Q11) and land ambience (Q59). Save each under a NEW file name and drag it onto the cue.

Sound effects — R13.1 pack

38 new *_v2 files built from the 40-sound SFX pack, plus Q19.2 rain burst and Q57.5 voice-restore resolve. Beds loop (Q4, Q20, Q22, Q31, Q45, Q50, Q60); Q25 runs 60 s and Q26 90 s. Full map: Part E §6.

R-06 • R-09 • R-19

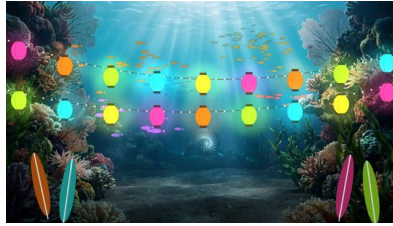
Legal, coordinated wireless frequencies (no 694–820 MHz). 8 radio packs + 1 spare; consider packs for the two Spirits. Check levels with an SPL meter for a family audience.

9 Projection and media

Backdrops R10 in the R13.1 cue plan (Part I). Each picture change is a 2 s crossfade in the same GO as its light and sound; blackouts fade to black (plus projector AV-mute).



Q4 • BG-02 Beneath the Waves — the coral



Q8 • BG-22 Rock Lobster beach party



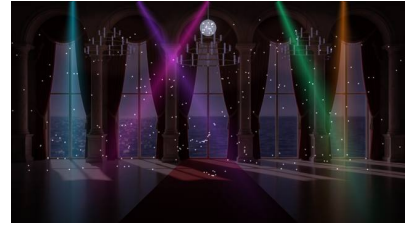
Q20 • BG-07 The storm



Q27 • BG-21 After the storm — clearing sky



Q39 • BG-18 Grand ballroom



Q40 • BG-23 Ballroom rave



Q50 • BG-19 Dingy cave with jars and bones



Q60 • BG-20 Palace gardens wedding (outdoors)



Q64 • BG-26 Tall ship at sunset

R-11 • R-22

Confirm throw and screen fill, treat the strip between projector and screen as a shadow zone, run HDBaseT or active HDMI from FOH, and rate the bar and bond the projector.

10 Haze, mirror ball and atmosphere

Cue	When	Action	Master cue
HZ-01	Before opening	Short burst	03–04
HZ-02	Opening / underwater	Low haze	04–08
HZ-03	Before clear comedy	Stop / clear	09
HZ-04	Storm standby	Short burst	20
HZ-05	Storm peak	Controlled burst	25
HZ-06	Storm end	OFF	27
HZ-07	Magic standby	Short burst	33–34
HZ-08	Octavia's lair	Low haze	50
HZ-09	Voice transfer	Reduce / OFF	57
HZ-10	Finale (optional)	Low only	63

Haze = hazer #40 on Mantra **P5 M4** (50 % output, fan 50 %), run on its fader 20–40 s before the look. Mirror ball only with a rated motor and bond (R-14); pinspots #41–42 if used.

R-04 • FLASH AND HAZE NOTICE

Single white hits only (Q21, Q23, Q25, Q35, Q57b), never above 3 a second. Post the printed notice (docs/FOH_Flashing_Lights_and_Haze_Notice.pdf) at the doors and add a line to the programme. Written haze plan with the venue (R-13).

11 Stage, props and practical effects

Props list and presets: Part H. Water pistols at Q19.2 and Q58 spray the audience (R-23): clean water, small amounts, below head height, away from electronics and hearing aids, warning before the show.

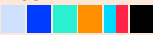
Cue	Change	Controls (R-12)
17	Scene Two → Ship	Glow-tape marks; crew in blacks; SM confirms clear before Q18
37	Act One end	Tabs close first; house up only after the stage is clear
44	Palace → Jellyfish	Cue cards struck; jellyfish preset
49	Jellyfish → Lair	Shell plinth onto the SP4 glow mark; HZ-08 standby
58.5	Lair → Dry land	Towels, dry, strike plinth

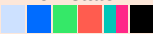
12 Master cue run — 83 cues + 10 songs

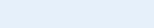
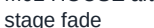
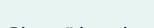
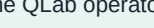
One shared sequence for the prompt book and all operators, read from the R13.1 workspace. Page = GO page in the licensed script as supplied. Tinted = critical; blue = song (its section GOs follow the music).

Swatch order: **Face** (FOH #1–6) · **Top** (LX1 #13–18) · **Side** (booms) · **Back** (LX2 COB) · **Pix** (PixBars) · **Spec** (specials). Colour as programmed in the R13.1 show file, darker = lower level, black = off.

Q	Moment	Trigger	LX / Mantra	QLab sound · video	FX / note	Page
1	Preshow preset	House opens	M02 HOUSE 	BG-01 · House music (placeholder)	Haze OFF; operators at preset	
2	House open	Audience admitted	M02 HOUSE 	Preshow music (placeholder)	Stage sealed / props preset	
3	Places warning	SM calls places	M02; stage preset ready 	—	HZ-01 standby	
4	Show start	House out / opening ready	M03 UNDERWATER 	BG-02 · low ocean bed	Short haze pre-burst	p 6
5	Spirit entrance	Spirit appears	SP2 + M03 	Spirit magic SFX	Low haze	p 6
6	Octavia entrance	Octavia appears / boos	SP3 + darker M03 	Evil chord	Hold atmosphere	p 6
7	Octavia exit	Exit beat	SP3 down / M03 	—	No extra effect	p 6
8	Opening number	Spirit exits / music starts	M03 UNDERWATER 	BG-22	Ensemble vocal mix	p 7
S1	Rock Lobster	Straight after Q8	P4 M1 · 7 sections	S01 Rock Lobster (original).mp3	supplied	
9	Theodore entrance	Character entrance	M04 COMEDY 	BG-17 · Optional short sting	Clear haze	p 8
10	Queen Marina entrance	Marina entrance	M05 ROYAL accent 	Fanfare	Return to underwater/royal balance	p 9
11	Ariel upset / exit	Emotional shift	M03 softer / SP1 if required 	Optional soft resolve (placeholder)	Keep projection calm	p 12
12	Singing Lesson preset	Scene Two begins	M04 COMEDY 	BG-03 · Optional comic intro	Audience interaction setup	p 12
13	Audience interaction	House lights up as scripted	M04 + house 15-20% 	—	Do not over-light audience	p 13
14	Dame song	Dame starts song	M04 COMEDY 	—	Dame mic priority	p 14
S2	Feeling Good (Austin Millz Remix)	Straight after Q14	P4 M2 · 6 sections	S02 Feeling Good (vocal - swap for backing).mp3	vocal version - swap for backing	
15	Ariel solo	Ariel starts solo	SP1 + M03 	—	Ariel mic priority	p 16
S3	Part of Your World	Straight after Q15	P4 M3 · 7 sections	S03 Part of Your World (vocal - swap for backing).mp3	vocal version - swap for backing	
15.5	Ship above - Ariel looks up	FLANDERS: "That was incredible."	Lighting shifts: SP1 + cool top-light shaft 	—	Ariel looks up	p 16

Q	Moment	Trigger	LX / Mantra	QLab sound · video	FX / note	Page
16	Octavia entrance	Octavia appears	SP3 / purple-dark accent 	Evil chord	Optional low haze	p 16
16.5	Octavia conjures the storm	"...this is going to be spectacular!"	Lighting churns: M11 + slow swirl 	BG-04 · Magic / villain swell	Optional short haze	p 17
17	Scene Two blackout	Scene ends	M01 BLACKOUT 	—	House remains independent	p 17
18	Ship number	Ship scene / song starts	M06 SHIP 	BG-05	Optional low haze	p 19
S4	Wellerman (Sea Shanty)	Straight after Q18	P4 M4 · 7 sections	S04 Wellerman (swap for backing if vocal).mp3	check for vocals	
19	Weather gag	Weather report beat	M06 + quick accent 	BG-06 · Wind / comedy FX / rain cue as scripted	Wet FX standby only	p 20
19.1	Weather: gusts of wind	"With gusts of wind"	-	Fart sound	Comedy beat	p 20
19.2	Weather: periodic rain	"And periodic rain."	-	Periodic rain burst	Water pistols spray audience (R-23)	p 20
19.3	Weather: back to normal	Pistols stop	Return to M06 SHIP 	BG-05	Towels ready	p 20
20	Storm wind build	Storm begins	M08 STORM 	BG-07 · Wind + rain build	HZ-04 burst	p 21
21	Lightning 1	First lightning action	M09 LIGHTNING -> M08 	Lightning crack + thunder	Short flash only	p 21
22	Storm build	Action escalates	M08 STORM 	Wind/rain + ship creaks	Controlled haze	p 21
23	Major lightning	Major storm beat	M09 LIGHTNING -> M08 	Close thunder / impact	Protect dialogue headroom	p 22
24	Ship lurch / water	Visible ship movement / spray	M08 with intensity shift 	Ship groan + water impact	Approved water effect	p 22
25	Storm peak	Maximum storm action	M09 / M08 	Major thunder + heavy rain	HZ-05 short controlled burst	p 22
26	Evan overboard	Evan disappears / search	Low blue storm tail 	Splash + low water bed	No flash unless scripted	p 22
27	Storm stop / shore transition	Storm visibly ends	M06 -> M07 SHORE 	BG-21	HZ-06 OFF	p 22
27.5	Scene Three end	"Then it's still my royal prerogative."	Tabs close; fade to preset 	—	Deck: sweep salt, dry check	p 23
28	Spirit on shore	Spirit appears	SP2 + M07 	BG-08 · Spirit magic	Clear stage	p 24
28.5	Spirit disappears	"...next for this prince and princess."	SP2 out; M07 holds 	Magical sound	-	p 24
29	Evan wakes	Evan opens eyes	Evan/area special or warm lift 	Small shimmer then silence	Dialogue priority	p 24
30	Ariel/Evan duet	Duet begins	M07 + SP1 	—	Two lead mics priority	p 25
S5	(I've Had) The Time of My Life	Straight after Q30	P4 M5 · 8 sections	S05 The Time of My Life (vocal - swap for backing).mp3	vocal version - swap for backing	
30.5	Scene Four end	Evan and Willis follow off	BLACKOUT 	—	Deck change to Scene Five	p 26
31	Underwater bar / Scene Five	Return underwater	M03 richer colour 	BG-02 · Ocean/bar ambience	Optional low haze	p 27
32	Octavia bar entrance	Octavia enters	SP3 / dramatic accent 	Evil chord	Hold low haze if used	p 29
33	Bargain standby	Ariel agrees / hand about to be taken	M10 ready 	BG-09 · Magic build ready	HZ-07 standby	p 30
34	Transformation build	Hands meet / spell starts	M10 MAGIC 	Magic swell / build	Short haze burst	p 30

Q	Moment	Trigger	LX / Mantra	QLab sound · video	FX / note	Page
35	Transformation flash	Transformation completes	Brief M09-like white hit -> new state 	Transformation impact	No repeated flash	p 31
36	Voice into shell	Voice moves / shell activates	SP1 -> SP4 / M12 	Shell / voice SFX	Keep shell brightest point	p 31
36.5	Octavia exits; court enters	"...evil plans don't make themselves!"	SP3 out; back to M03 underwater 	BG-02 · Evil chord	-	p 32
36.7	Spirit enters	Theodore panics - after a moment	SP2 + M10 accent 	Magical sound	-	p 32
37	Act One end	Tabs close	M01 BLACKOUT 	End-act / interval music (placeholder)	House up after clear	p 33
38	Interval	House open	M02 HOUSE 	BG-16 · Interval music (placeholder)	Reset / dry / check	
39	Act Two start	House out / palace preset	M05 ROYAL 	BG-18 · palace intro	Haze OFF	p 34
40	Palace opening song	Opening number starts	M05 ROYAL 	BG-23	Lead/ensemble mix	p 34
S6	Crab Rave	Straight after Q40	P4 M6 · 7 sections	S06 Crab Rave (original).mp3	supplied	
41	Evan memory beat	Quiet memory beat	General 45% + focused warm area 	BG-18 · Optional very low underscore	No effect	p 34
42	Octavia enters palace	Evil chord, Octavia enters	Royal drains to purple/dark 	Evil chord / tension bed	Dialogue priority	p 36
42.5	Octavia sings with Ariel's voice	Octavia holds up the shell	SP3 warm + shell glint 	Duet line in Ariel's voice (placeholder)	Playback or offstage mic	p 36
43	Cue-card routine	"Its Coral singers"	Readable soft palace look 	Cue-card underscore (placeholder)	Keep cards visible	p 37
44	Palace scene end	Blackout / transition	M01 BLACKOUT 	—	Prepare underwater	p 38
45	Jellyfish search	Search scene begins	M13 JELLYFISH 	BG-11 · comic/ocean bed	Optional low haze	p 39
45.5	Curtains open - jellyfish	"Not that kind of sting."	Full M13 JELLYFISH as tabs open 	—	Tabs open	p 40
46	Jellyfish chorus	After "After four... Four!"	Hold M13 	—	Audience-interaction timing	p 41
S7	Jellyfish chorus (song to be decided)	Straight after Q46	P4 M7 · 4 sections	S07 song TBA.wav	song still to choose (silent placeholder)	
47	Jellyfish gag / repeats	Jellyfish appears / repeats	Cyan accent / pulse then M13 	Short jellyfish sting / resume chorus	Manual repeat as scene requires	p 41
48	Spirit returns	Spirit appears	SP2 + M10 accent 	Spirit magic	Short optional haze	p 42
49	Scene blackout	Spirit exit / scene end	M01 BLACKOUT 	—	Lair ready	p 42
50	Lair reveal	Octavia lair established	M11 LAIR + SP3/SP4 	BG-19 · dark ambience	HZ-08 low	p 42
51	Octavia song	Song begins	M11 LAIR 	—	Octavia/ensemble mix	p 42
S8	Poor Unfortunate Souls	Straight after Q51	P4 M8 · 7 sections	S08 Poor Unfortunate Souls (original).mp3	supplied	
52	Shell activation	Shell opened / activated	M12 VOICE SHELL 	BG-09 · Shell magic	Low haze; shell isolated	p 44
53	Voice transfer - Dame	Voice passes to Dame	Dame fixed special up; shell down 	Transfer sting	Actor must hit mark	p 44

Q	Moment	Trigger	LX / Mantra	QLab sound · video	FX / note	Page
54	Voice transfer - Flanders	Voice passes to Flanders	Crossfade Dame -> Flanders 	Transfer sting	Manual actor-paced cue	p 45
55	Voice transfer - Theodore	Voice passes to Theodore	Crossfade Flanders -> Theodore 	Transfer sting	Manual actor-paced cue	p 45
56	Voice transfer - Marina	Voice passes to Marina	Crossfade Theodore -> Marina 	Transfer sting	Manual actor-paced cue	p 45
57	Haywire	Voice chaos resolves to Ariel	Fast specials -> brief flash -> SP1 	Magic sequence + resolve	HZ-09 reduce/OFF	p 46
57b	Ariel restored (57b > 57c)	Voice chaos resolves to Ariel	Fast specials -> brief flash -> SP1 	Voice restore / resolve	HZ-09 reduce/OFF	p 46
58	Cross-fire exit	Music starts / characters exit	Fast bright transition 	BG-19 · Chase/comedy music	Water pistol FX only in approved zone	p 47
58.5	Scene Eight end	Stage clear after cross-fire	Tabs close; fade 	—	Deck: towels, dry, strike plinth	p 47
59	Dry-land reveal	Next land scene established	M14 DRYLAND 	BG-13 · optional land ambience (placeholder)	Stage dry check	p 47
59.5	Spirit appears to Evan	Willis exits; Evan alone	SP2 + dim M14 	Magical sound	-	p 49
59.7	Scene Nine end	"...the voice that sings to your heart!"	Dim, then BLACKOUT 	Magical sound	Deck change to wedding	p 49
60	Wedding preset	Wedding palace ready	M15 WEDDING 	BG-20 · optional ceremony bed	Optional low haze only if desired	p 50
61	Octavia entrance	Octavia interrupts wedding	SP3 + villain shift 	Evil chord	Haze OFF	p 51
61.5	Stop the wedding!	"...speak now or..."	Bump M15 accent 	Comic sting	-	p 51
62	Spirit / reveal / arrest beats	Positive reveal begins	M10/SP2 -> M15 	Magic sting / dialogue support	No effect that masks speech	p 54
63	Full-cast wedding song	Final Spirit speech resolves into song	M15 -> celebratory build 	BG-25	Optional low haze; full-cast balance	p 54
S9	Absolutely Everybody	Straight after Q63	P4 M9 · 7 sections	S09 Absolutely Everybody (original).mp3	supplied	
63.5	Scene Ten end	End of song	Tabs close; fade 	—	Bows preset	p 54
64	Bows	Cast line-up / bow starts	M16 BOWS; optional house 15% 	BG-26	Keep stage evenly lit	p 54
S10	He's a Pirate	Straight after Q64	P4 M10 · 6 sections	S10 Hes a Pirate (original).mp3	supplied	
65	End / exit	Final applause settles	M02 HOUSE after stage fade 	Exit music (placeholder)	Post-show reset begins	p 55

R-01 · R-31

The contents page lists “Scene Eleven: On the Shore” but the text runs Scene Ten (pp 51–54) straight to Scene Twelve (p 55); the Scene Five song listed on p 33 is also missing. Confirm with the director and licensor before cue lock.

Agree at the paper tech whether the DSM or the QLab operator calls the song-section GOs.

13 Critical-sequence cards

Sequence	Step	Lighting (Mantra)	Sound / video (QLab)	Haze / practical
Storm Q20–27	20	P2 M3 c4 M08 storm	Wind/rain loop · BG-07	HZ-04 burst
	21 / 23 / 25	White hit → return +0.2 s	Crack + thunder	Short flash only
	22 / 24	Storm build / ship lurch	Creaks · groan + water	Water only if approved
	26 / 27	Low tail → shore	Splash + bed · storm layers out · BG-21	HZ-06 off; dry stage
Transformation Q33–36	33 → 36	M10 build · white hit → 35b (+0.3 s) · SP1 → SP4	Magic build · impact · shell voice · BG-09	HZ-07
Voice transfer Q52–57b	52 → 56	Shell → V1 → V2 → V3 → V4 (0.3 s)	Shell magic · transfer sting ×4	Actors on marks
	57 / 57b	HAYWIRE chase on · restore hit, chase off, SP1 (+0.3 s)	Haywire SFX · restore resolve	HZ-09 reduce/off
Jellyfish Q45–49	45 → 47	M13 · pulse → return +1 s · repeats on P5 M6	Comic bed · S7 chorus · sting	Performer sightline

RECOVERY

Storm: stay in the storm and wait for the next action. Transformation: continue with the new state; never repeat the hit. Voice: go straight to the next correct special. HAYWIRE keeps running until Q57b fires.

14 Rehearsal and technical week

Session	Focus	Result
Paper tech	SM + operators walk all cues and song sections	Every cue has a trigger, page and owner (R-15, R-31)
Tech 1	Hang, patch, focus	All fixtures respond; specials focused; looks checked from the house (R-30)
Tech 2	QLab / projection	Show-Mac settings set (R-29); every backdrop on the real screen
Tech 3	StudioLive / mics	Casting-based mic plan; spare tested; scan done
Dry tech	Storm, transformation, voice transfer + HAYWIRE	Timings proven with stand-ins
Tech 4 / 5	Cue-to-cue Act One / Act Two	Flash returns, releases, song fades, Q57 → Q57b
Wet tech	Water + haze + projector	Effects safe with final cable positions
Dress 1–2	Full runs	Timing lock; photos of looks (R-20)

15 Pre-show, interval and post-show

Time	Action
T-90	Power up in order (Quick Reference card 7); Mantra loads the Default Show
T-80	QLab Mac: open the show from the Desktop folder; no red cues; Stream Deck profile
T-70	E3 SAFE LIGHT → desk responds; test tone on ch 11–12; Q4 backdrop; E2 black
T-60	Storm, transformation, voice-transfer checks (low level, haze off)
T-45	Panic, O (All Cues Off), playhead on Q1, Show Mode; props, shell, cue cards, water preset
T-30	House open: SM calls Q1
☐ Interval: dry water areas; refill pistols for Q58 ☐ Interval: QLab on Q39; desk ready on P3 ☐ Post-show: projector cool-down ☐ Interval: haze fluid; batteries ☐ Post-show: Panic, save, back up any change ☐ Post-show: dry props, log faults	

16 Emergency, recovery and troubleshooting

Failure	Immediate action	Continue?
QLab freezes	E1/Panic if possible; lighting from the desk memories; emergency playback ch 13–14 on SM call	Yes
QLab → Mantra link fails	Run P2/P3/P4 memories with Next Cue (Next Cue again after each flash)	Yes
Looks stacking / wrong colour	O (All Cues Off) then re-fire the current cue	Yes
Radio mic dies	Mute the channel; spare pack at the next exit	Yes
Projector fails	E2 black / AV-mute; continue on lighting	Yes
Water or haze unsafe	Stop it; continue the dry version	Yes
Fire alarm / evacuation	HOLD; house lights full; E1 stop; venue procedure	Venue decides
Recovery state	Where	
SAFE LIGHT	QLab E3 · Mantra P2 M10 (also P7 M7 = M17)	
Projector black	QLab E2 · projector AV-mute	
Stop playback	QLab E1 / Esc — live mics continue	
Emergency playback	Backup device on StudioLive ch 13–14 (R-08)	

17 Backup, version control and handover

Asset	Method	When
Show folder (TLM_R13_REBUILT_Show_Files)	Dated copy of the whole folder on a USB SSD and a second Mac	After every change session and at lock
Mantra show	Desk + USB export	After patch, cue-to-cue and lock
StudioLive scenes	Save baseline scene TLM SHOW	After sound lock
Cue sheets / book	PDF + printed copies	After cue lock

Checksums of the checked QLab and Mantra files are in BUILD_METADATA.json. Only one copy is LIVE; record every post-lock change in the change log.

Date/time	System	Change	Reason	Tested by	Approved

18 Equipment checklist

Category	Item	Qty	Status	✓
Lighting	LSC Mantra Lite + 2 wings	1	Venue	☐
Lighting	Lightsky C42 / Zoom / TourCOB / PixBar / hazer	12/10/11/6/1	Venue — confirm #38 and #39 exist	☐
Rigging	Floor booms SR + SL with bases, arms, bonds, sandbags	2	R-26	☐
Control	Art-Net/sACN node, show switch, labelled Cat6	1 each	R-17	☐
Playback	Mac with QLab 5 + licence	1	R-07, R-21	☐
Playback	Emergency playback device	1	R-08	☐
Sound	StudioLive 16, 8 radio packs + spare, area mics	—	R-06, R-09	☐
Projection	Epson projector + HDBaseT	1	R-11, R-22	☐
Show control	Stream Deck 15-key	1	Part L	☐
Safety	Glow tape, running lights, towels, squeegee, non-slip mat	—	R-12	☐

19 Final technical sign-off

- ☐ Licensed script and prompt copy confirmed
- ☐ Show-Mac QLab settings set (R-29)
- ☐ SAFE LIGHT from QLab and desk
- ☐ Flash returns and Q57 → Q57b checked
- ☐ Backing tracks and placeholders replaced (R-28)
- ☐ Backdrops checked on the real screen
- ☐ Wireless scanned; spare tested
- ☐ Water and haze rehearsed
- ☐ Test-and-tag / RCD confirmed
- ☐ Cue numbers locked in every system
- ☐ Mantra OSC trigger added
- ☐ Every P2/P3/P4 memory fired from QLab
- ☐ Song fades and releases checked
- ☐ S7 jellyfish song chosen
- ☐ Looks checked from the house (R-30)
- ☐ Emergency playback tested
- ☐ FOH notice posted
- ☐ All show files backed up

Role	Name	Signature	Date
Technical Director			
Stage Manager			
Show operator			
Sound			
Deck / ASM			
Venue contact			

Appendices — worksheets

A Focus and position worksheet

#	Position	Focus / mark	Colour / level notes
#1	FOH		
#2	FOH		
#3	FOH		
#4	FOH		
#5	FOH		
#6	FOH		
#7	FOH		
#8	FOH		
#9	LX1		
#10	LX1		
#11	LX1		
#12	LX1		
#13	LX1		
#14	LX1		
#15	LX1		
#16	LX1		
#17	LX1		
#18	LX1		
#19	LX2		
#20	LX2		
#21	SR boom		
#22	SL boom		
#23	LX2		
#24	LX2		
#25	LX2		
#26	LX2		
#27	LX2		
#28	LX2		
#29	SR boom		
#30	SR boom		
#31	SL boom		
#32	SL boom		
#33	LX2		

#	Position	Focus / mark	Colour / level notes
#34	LX2		
#35	LX2		
#36	LX2		
#37	LX2		
#38	LX2		
#39	LX2		

B Operator show report

Cue / time	Department	Issue	Action taken	Owner

C Sources

- Licensed script, Nick Lawrence Pantomimes, revised April 2026 (as supplied); casting sheet 28 Sep 2026.
- R13.1 show files: TLM_Show_R13_1.qlab5, TLM_SHOW_2026_R13_FLASHY_SCENE_SPLIT.mtr, section map, media manifest, SFX retarget map, fix list.
- Production Book R8.1, R13 User Guide, R13 Production Book update, R13.1 review and fix list, Backdrops R10.
- LSC Mantra Lite User Manual v3; Figure 53 QLab 5 documentation.

Technical notes